Spatio-Temporal and Spatial Econometric Analysis of Housing Prices: Clustering, Heterogeneity, and Model Validation

Sasank Lingareddy(IMT2023120), Akul Anhith(IMT2023558), Thrilochan Reddy(IMT2023527)
Department of Computer Science and Engineering
International Institute of Information Technology, Bangalore
Bangalore, India
{sasank.reddy, motukuri.akul, thrilochan.reddy}@iiitb.ac.in

***Abstract*—This study presents a comprehensive spatio-temporal analysis of housing prices using spatial statistical and econometric techniques. Feature engineering was performed to construct meaningful spatial and temporal variables, including building age, renovation status, and an accessibility index derived from proximity to schools and airport distance. Global spatial autocorrelation analysis using Moran's I revealed strong and statistically significant clustering in housing prices (I = 0.721, p = 0.001) and building age (I = 0.495, p = 0.001), confirming the presence of spatial dependence. Local Indicators of Spatial Association (LISA) further identified dominant High–High and Low–Low clusters, indicating clear spatial market segmentation.**

Unsupervised spatial clustering using K-means demonstrated optimal segmentation at 3–4 clusters based on silhouette scores, capturing distinct housing submarkets. To examine spatial heterogeneity, Ordinary Least Squares (OLS) regression was compared with Geographically Weighted Regression (GWR). While OLS explained 63.5% of the variance (R^2 = 0.635), GWR significantly improved model performance (R^2 = 0.916), highlighting substantial spatial non-stationarity in coefficient estimates. Spatial holdout validation further compared OLS, Spatial Lag, and Spatial Error models, demonstrating improved predictive stability under spatially explicit modeling frameworks.

Overall, the results confirm strong spatial autocorrelation, localized heterogeneity in housing determinants, and the importance of spatial econometric modeling for accurate price estimation and market interpretation.

***Index Terms*—Spatio econometric analysis, housing prices, clustering.**

I. Introduction

Housing markets are spatially structured systems in which property values are influenced by structural characteristics, neighborhood context, and geographic location, consistent with Tobler's First Law of Geography [1]. Prices often cluster due to shared amenities, infrastructure access, and localized demand patterns. Temporal factors such as construction period and renovation activity further shape valuation trends. Traditional regression models assume spatial independence and global parameter stability; however, when spatial autocorrelation and heterogeneity exist, these assumptions may lead to biased or inefficient results.

This study performs a comprehensive Spatio-Temporal Data Analysis (STDA) of housing prices by integrating feature engineering, spatial statistics, clustering, and spatial econometric modeling. Derived variables include log-transformed price, building age, a renovation indicator, and an accessibility index based on proximity to schools and airport distance. These features capture structural depreciation effects and locational advantages.

Global spatial autocorrelation analysis using Moran's I revealed strong and statistically significant clustering for housing prices ($I \approx 0.721$, $p = 0.001$) and building age ($I \approx 0.495$, $p = 0.001$), while accessibility showed no significant clustering. Local Indicators of Spatial Association (LISA) identified dominant High–High and Low–Low clusters, indicating clear spatial segmentation in the housing market. K-means clustering further supported the presence of 3–4 distinct market segments based on price, location, and living area.

To examine spatial heterogeneity, Ordinary Least Squares (OLS) was estimated as a baseline (R^2 = 0.635), followed by Geographically Weighted Regression (GWR), which significantly improved model performance (R^2 = 0.916). The results indicate strong spatial non-stationarity in price determinants, particularly for accessibility. Spatial holdout validation comparing OLS, Spatial Lag, and Spatial Error models further evaluated predictive robustness under spatial dependence.

By integrating spatial diagnostics, clustering, and geographically adaptive regression within a unified framework, this study provides both explanatory and predictive insights into housing market dynamics. The findings reinforce that spatial dependence and non-stationarity are not peripheral effects but central characteristics of real estate data, necessitating spatially explicit modeling approaches for reliable inference and market interpretation.

II. Data Description

The dataset consists of **14,619 residential property observations**, each associated with structural, locational, and temporal attributes. The dependent variable is housing price, which was log-transformed using $\log(1 + \text{Price})$ to stabilize variance and reduce right-skewness commonly observed in real estate data.

A. Structural Attributes

Key structural variables include:

- Living area (continuous, square footage)
- Number of bedrooms (discrete count)
- Grade of the house (ordinal quality indicator)
- Built Year, used to derive building age
- Renovation Year, used to construct a binary renovation indicator

Building age was computed as:

$$\text{building_age} = 2026 - \text{Built Year}$$

Renovation status was encoded as:

$$\text{renovated} = \begin{cases} 1 & \text{if Renovation Year} > 0 \\ 0 & \text{otherwise} \end{cases}$$

B. Locational Attributes

Spatial information is embedded through geographic coordinates, enabling spatial weights construction and neighborhood structure modeling. Additionally, proximity-based variables were included:

- Number of schools nearby
- Distance from the airport

These were combined into an accessibility index defined as:

$$\text{accessibility_index} = \frac{\text{Number of schools nearby}}{\text{Distance from the airport} + 1}$$

The denominator adjustment avoids division by zero and ensures numerical stability.

C. Temporal Attributes

Temporal dynamics are captured through:

- Construction decade (derived from Built Year)
- Renovation year trends
- Building age as a continuous temporal decay proxy

These variables allow examination of price evolution across historical development phases and renovation cycles.

D. Spatial Structure

A distance-based spatial weights matrix was constructed to model neighborhood relationships following standard spatial statistical methodology [2]. During construction, the weights matrix exhibited multiple disconnected components, indicating that certain properties form isolated spatial subgraphs. This characteristic is relevant for interpreting spatial autocorrelation and spatial regression results.

Overall, the dataset integrates structural, spatial, and temporal dimensions, making it suitable for comprehensive spatio-temporal statistical modeling and spatial econometric analysis.

III. Exploratory Spatial and Clustering Analysis

This section examines the presence of spatial structure, dependence, and market segmentation in the housing dataset prior to formal econometric modeling.

A. Spatial Weights Construction

A distance-based spatial weights matrix was constructed to model neighborhood relationships among properties. The weights matrix defines spatial interaction based on geographic proximity. During construction, a warning indicated multiple disconnected components in the spatial graph, implying that certain subsets of properties form isolated spatial clusters. This structural characteristic is important for interpreting spatial autocorrelation and regression results, as it suggests localized submarkets with limited spatial interaction.

B. Global Spatial Autocorrelation

Global spatial autocorrelation was assessed using Moran's I statistic [3]. The results indicate strong and statistically significant clustering for housing prices:

$$I_{\text{price}} \approx 0.721, \quad p = 0.001$$

This confirms that similar price values are spatially concentrated rather than randomly distributed (see Fig. 1).

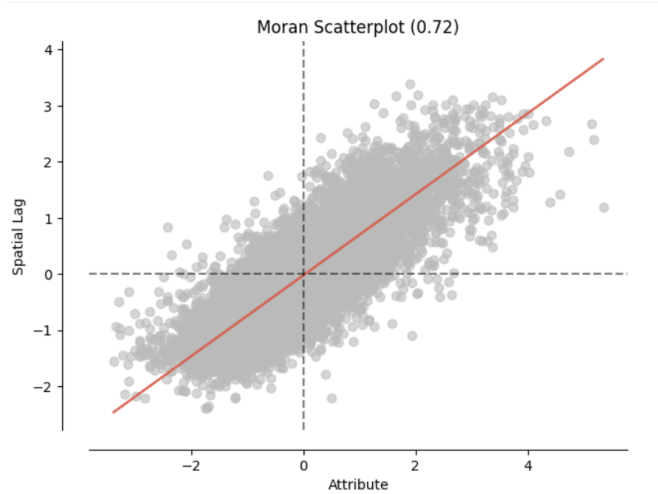

Fig. 1. Moran scatterplot for log-transformed housing prices. The strong positive slope confirms significant positive spatial autocorrelation ($I \approx 0.721$, $p = 0.001$).

Building age also exhibits moderate and significant clustering (see Fig. 2):

$$I_{\text{age}} \approx 0.495, \quad p = 0.001$$

In contrast, the accessibility index does not show significant global spatial autocorrelation:

$$I_{\text{accessibility}} \approx -0.004, \quad p = 0.152$$

These results suggest that price and structural aging patterns are spatially dependent, whereas accessibility is more spatially dispersed (see Fig. 3), consistent with distance-based spatial association theory [4].

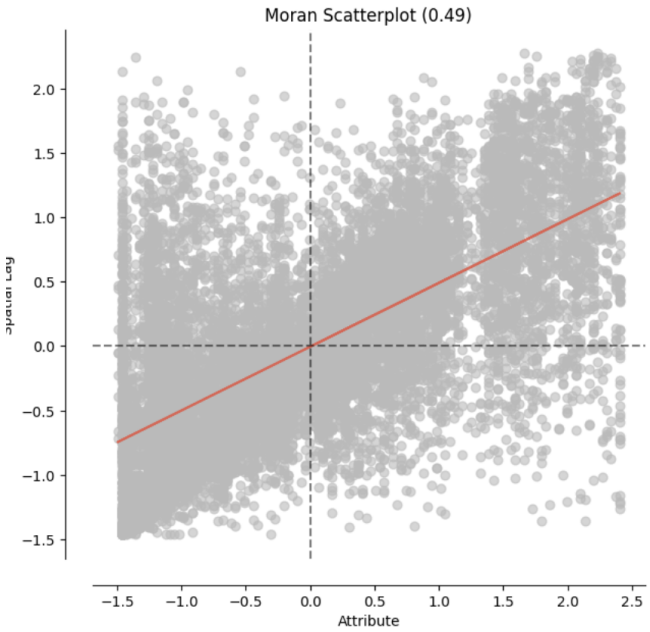

Fig. 2. Moran scatterplot for building age. A positive slope indicates moderate spatial clustering ($I \approx 0.495$, $p = 0.001$).

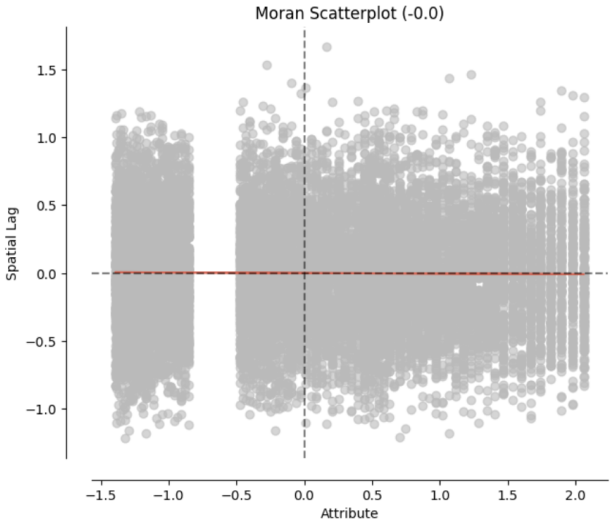

Fig. 3. Moran scatterplot for accessibility index. The near-zero slope indicates absence of significant global spatial autocorrelation ($I \approx -0.004$, $p = 0.152$).

C. Local Spatial Autocorrelation (LISA)

Local Indicators of Spatial Association (LISA) were computed to identify localized clustering patterns [5]. Significant clusters were categorized into High–High (HH), Low–Low (LL), High–Low (HL), and Low–High (LH) types. The majority of significant observations belonged to High–High and Low–Low clusters, indicating strong spatial concentration of both high-value and low-value housing areas. This confirms the presence of distinct spatial submarkets (see Fig. 4).

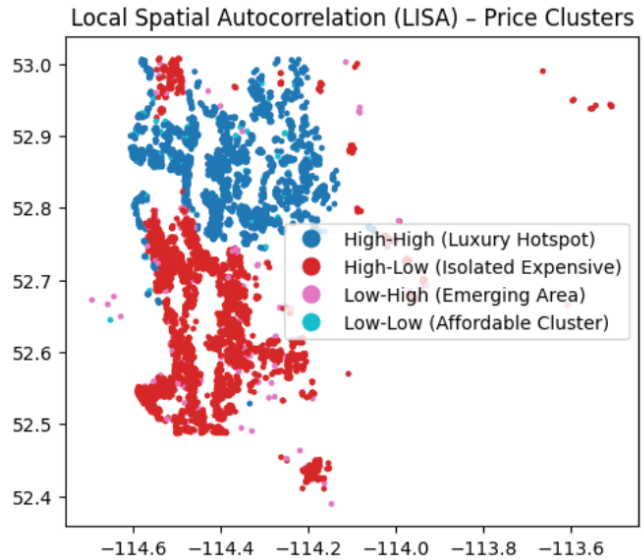

Fig. 4. LISA cluster map for log-transformed housing prices. High–High (HH) and Low–Low (LL) clusters dominate, indicating strong localized spatial concentration of similar price values.

D. Spatial Clustering (K-Means)

To further identify market segmentation, K-means clustering was applied using spatial location, $\log(1 + \text{Price})$, and living area. The optimal number of clusters was evaluated using silhouette scores for $K = 2$ to 9. The highest silhouette values were observed for K = 3 and K = 4, indicating well-separated cluster structures (see Fig. 5).

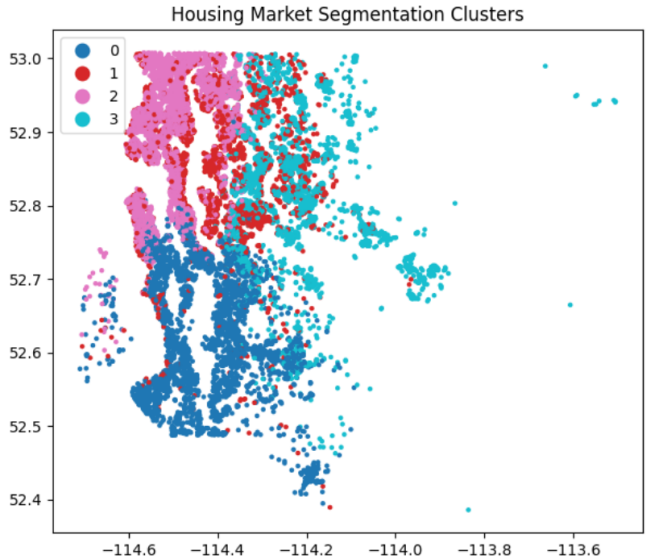

Fig. 5. Spatial K-means clustering with $K = 3$. The solution reveals three geographically distinct housing submarkets.

Final clustering solutions revealed clearly differentiated market segments with varying property counts, supporting the hypothesis of geographically structured housing submarkets.

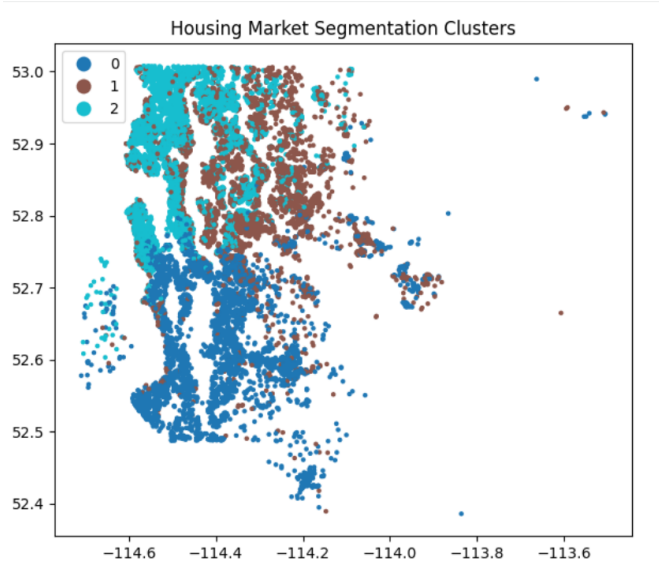

Fig. 6. Spatial K-means clustering with $K = 4$. Increased segmentation captures finer submarket differentiation.

These clusters reflect variation in price levels and structural characteristics across spatial regions (see Fig. 6).

Overall, the exploratory analysis confirms strong spatial dependence in housing prices, localized clustering behavior, and the existence of segmented market regimes, thereby justifying the application of spatial econometric modeling in subsequent sections.

IV. Spatial Regression and Heterogeneity Analysis

This section evaluates the determinants of housing prices using global and spatially adaptive regression models to examine spatial dependence and parameter non-stationarity within a spatial econometric framework [6].

A. Ordinary Least Squares (OLS) Baseline Model

An Ordinary Least Squares (OLS) regression model was first estimated using $\log(1 + \text{Price})$ as the dependent variable and structural as well as locational predictors, including living area, number of bedrooms, house grade, building age, and accessibility index.

The OLS model achieved:

$$R^2 = 0.635, \quad \text{Adj. } R^2 = 0.634$$

All structural variables were statistically significant at conventional levels, while the accessibility index was not significant ($p = 0.514$). The condition number (1.71×10^5) suggests potential multicollinearity or numerical scaling issues, warranting cautious interpretation of coefficients.

Although the model explains a substantial portion of variance, the presence of strong spatial autocorrelation in prices indicates that a global stationary model may be insufficient.

B. Geographically Weighted Regression (GWR)

To account for spatial heterogeneity, a Geographically Weighted Regression (GWR) model was estimated [7] using an adaptive bisquare kernel with a bandwidth of 122 nearest neighbors. Unlike OLS, GWR allows coefficients to vary spatially.

The GWR model substantially improved explanatory power:

$$R^2 = 0.916, \quad \text{Adj. } R^2 = 0.903$$

The reduction in residual sum of squares and the improved AIC indicate superior model fit compared to OLS. The effective number of parameters increased due to spatial adaptation, reflecting localized estimation.

C. Coefficient Variability and Spatial Non-Stationarity

Summary statistics of GWR parameter estimates demonstrate considerable spatial variability across predictors. While living area shows minimal variation, bedrooms and house grade exhibit moderate spatial fluctuation. The accessibility index displays the highest standard deviation among coefficients, indicating strong spatial instability in its effect on price.

This variability confirms the presence of spatial non-stationarity, meaning the relationship between housing price and explanatory variables changes across geographic space [8]. Such behavior cannot be captured by a global OLS model.

Overall, the comparison between OLS and GWR provides clear evidence that housing price determinants vary spatially. The substantial improvement in R^2 and localized coefficient estimates validate the necessity of geographically adaptive modeling for accurately capturing housing market dynamics.

V. Spatio-Temporal Trend Analysis & Spatial Model Validation

This section examines macro-level temporal and spatial trends in housing prices and evaluates predictive performance under spatially explicit validation.

A. Spatio-Temporal Trend Analysis

1) Construction Decade Trend: Average housing prices were aggregated by construction decade to examine long-term temporal evolution. Early 20th century properties (1900–1930) exhibited moderate average prices, followed by a noticeable decline around the 1940s–1960s period. From the 1970s onward, a steady upward trend is observed, with properties built after 2000 showing substantially higher average prices (see Fig. 7). This pattern suggests modernization effects and increasing demand for newer constructions.

2) Renovation Trend: Properties renovated between 2011 and 2015 display consistently elevated average prices relative to older, non-renovated properties. This indicates that renovation contributes positively to valuation, though the magnitude varies across years, suggesting localized market sensitivity (see Fig. 8).

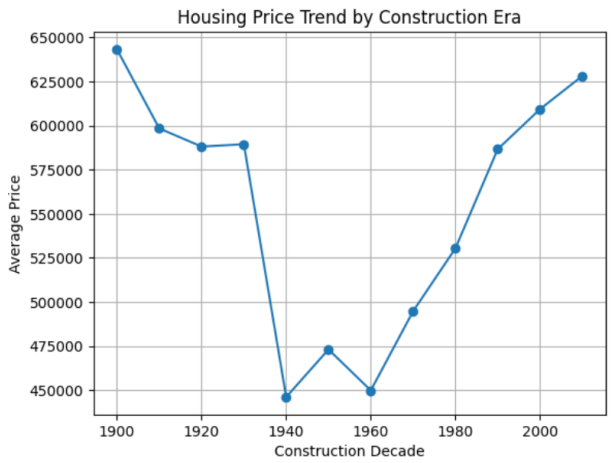

Fig. 7. Average housing price by construction decade. A clear upward trend is observed for properties built after 1970, with pronounced appreciation for post-2000 constructions.

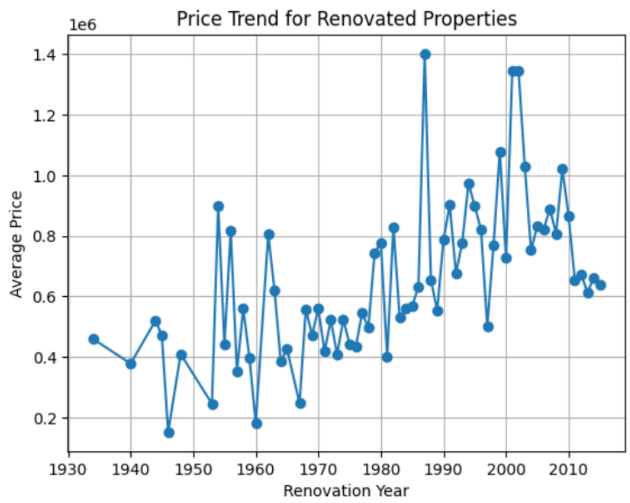

Fig. 8. Average housing price by renovation year. Recently renovated properties show consistently higher price levels compared to non-renovated properties.

3) Spatial Trend Gradient: A spatial trend surface was estimated using geographic coordinates. The trend coefficients indicate:

$$\beta_{\text{latitude}} \approx -0.143$$

$$\beta_{\text{longitude}} \approx 0.415$$

The positive longitudinal gradient suggests price expansion toward increasing longitude, while the negative latitudinal coefficient indicates a relative decline in the north–south direction (see Fig. 9). Together, these results confirm directional spatial price gradients across the study area (see Fig. 10).

B. Spatial Model Validation

To evaluate predictive robustness under spatial dependence, a spatial holdout validation approach was implemented. The

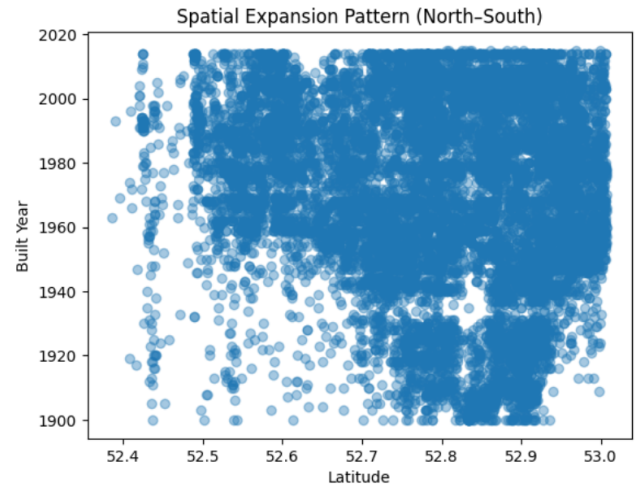

Fig. 9. North–South (latitude) spatial price gradient. The negative coefficient indicates relative price decline along the latitudinal direction.

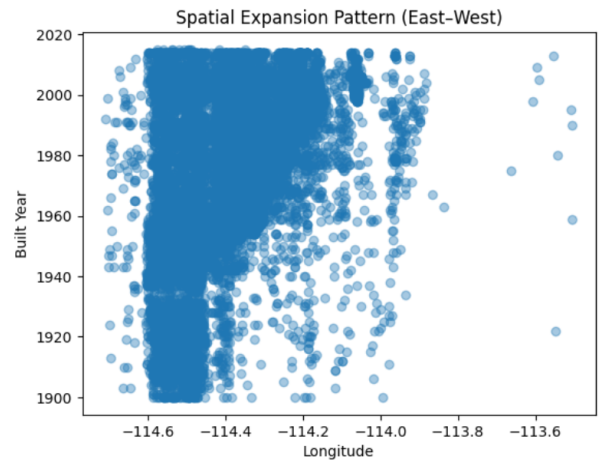

Fig. 10. East–West (longitude) spatial price gradient. The positive coefficient indicates price expansion toward increasing longitude.

dataset was divided into spatially structured training ($n = 7{,}311$) and testing ($n = 7{,}308$) subsets. Spatial weights were reconstructed for the training data, again revealing multiple disconnected components.

1) Model Comparison: Three models were compared using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):

- **Spatial Lag Model**

$$\text{RMSE} \approx 771{,}402, \quad \text{MAE} \approx 655{,}331$$

- **Spatial Error Model**

$$\text{RMSE} \approx 357{,}888, \quad \text{MAE} \approx 231{,}521$$

- **OLS (Baseline)**

$$\text{RMSE} \approx 373{,}483, \quad \text{MAE} \approx 209{,}783$$

The Spatial Error Model demonstrates substantially lower RMSE compared to the Spatial Lag Model, indicating that spatial dependence is more effectively captured through the error structure rather than a direct spatially lagged dependent variable. OLS performs comparably in MAE but slightly worse in RMSE relative to the Spatial Error Model.

Overall, validation results confirm the importance of accounting for spatial dependence in predictive modeling, while also highlighting that different spatial specifications capture distinct forms of spatial interaction.

VI. Results and Discussion

The empirical analysis provides strong evidence of spatial structure, market segmentation, and localized heterogeneity in housing prices.

A. Spatial Dependence and Market Segmentation

Global Moran's I confirmed significant positive spatial autocorrelation for housing prices ($I \approx 0.721$, $p = 0.001$), indicating that high-value and low-value properties are geographically clustered. Building age also exhibited moderate clustering, suggesting that development patterns are spatially structured. In contrast, accessibility did not show significant global clustering, implying spatial dispersion of locational advantages.

LISA analysis further revealed dominant High–High and Low–Low clusters, confirming the presence of concentrated high-price and low-price zones. K-means clustering supported this segmentation, with optimal solutions around $K = 3$ or 4 clusters. These findings collectively demonstrate the existence of geographically differentiated housing submarkets.

B. Spatial Heterogeneity in Price Determinants

The OLS model explained 63.5% of the variance in log-transformed prices, with structural variables such as living area, grade, and building age showing statistical significance. However, the substantial improvement observed in the GWR model ($R^2 = 0.916$) indicates that relationships between predictors and price vary significantly across space.

Coefficient variability analysis showed minimal spatial variation for living area, moderate variation for bedrooms and grade, and very high variability for accessibility. This confirms strong spatial non-stationarity, particularly in how locational accessibility influences property value. The results suggest that the effect of amenities and proximity is highly context-dependent rather than globally uniform.

C. Spatio-Temporal Dynamics

Trend analysis revealed increasing average prices for newer construction decades, reflecting modernization and market appreciation over time. Renovated properties also exhibited elevated price levels, reinforcing the value-enhancing role of property upgrades. The spatial gradient analysis indicated directional expansion of higher prices along the longitudinal axis, confirming systematic geographic price differentiation.

D. Predictive Performance of Spatial Models

Spatial holdout validation demonstrated that accounting for spatial dependence improves predictive stability. The Spatial Error Model achieved lower RMSE than the Spatial Lag Model, suggesting that unobserved spatially correlated factors play a more prominent role than direct spillover effects from neighboring prices. OLS produced competitive MAE values but slightly higher RMSE, indicating sensitivity to larger prediction errors.

Overall, the results confirm three central findings: (1) housing prices exhibit strong spatial clustering, (2) price determinants vary geographically, and (3) spatial econometric models provide improved explanatory and predictive performance compared to purely global approaches. These outcomes highlight the necessity of spatially explicit modeling frameworks in real estate analytics.

VII. Conclusion

This study conducted a comprehensive spatio-temporal analysis of housing prices by integrating spatial autocorrelation diagnostics, clustering techniques, spatial econometric modeling, and validation procedures. The results demonstrate that housing prices are not randomly distributed across space but exhibit strong and statistically significant spatial clustering. Both global and local spatial statistics confirmed the existence of geographically concentrated high-value and low-value submarkets.

Market segmentation analysis further revealed distinct housing clusters, supporting the presence of spatially differentiated price regimes. Regression modeling showed that while the global OLS model provides reasonable explanatory power, it fails to capture substantial spatial heterogeneity. The Geographically Weighted Regression (GWR) model significantly improved model fit, confirming strong spatial non-stationarity in housing price determinants, particularly for locational accessibility.

Spatio-temporal trend analysis indicated systematic price appreciation for newer construction decades and value enhancement associated with renovation activity. Spatial gradient estimation revealed directional expansion patterns, highlighting macro-level geographic price dynamics. Spatial holdout validation demonstrated that models accounting for spatial dependence, particularly the Spatial Error Model, provide improved predictive robustness compared to simpler global approaches.

Overall, the findings emphasize that spatial dependence and heterogeneity are fundamental characteristics of housing market data. Ignoring these properties can lead to incomplete inference and suboptimal prediction. The study underscores the importance of incorporating spatial statistical and econometric techniques in real estate analytics to achieve accurate modeling, interpretation, and decision support.

DECLARATIONS

Task Ownership

Akul: Data preprocessing for spatial analysis, including feature preparation and spatial weights construction. Conducted Global and Local Spatial Autocorrelation analysis using Moran's I and LISA, and performed spatial clustering based on LISA results.

Thrilochan: Spatial clustering using K-means with silhouette evaluation. Conducted spatial heterogeneity and stationarity analysis using Ordinary Least Squares (OLS) and Geographically Weighted Regression (GWR), including bandwidth selection and coefficient interpretation.

Sasank: Spatio-temporal trend analysis (construction decade trends, renovation trends, and spatial gradient analysis). Implemented spatial regression models (Spatial Lag and Spatial Error), performed spatial holdout validation, and conducted comparative analysis with standard OLS regression.

GenAI Tool Usage Declaration

ChatGPT (OpenAI) was utilized as an AI-based support tool during the course of this work.

1) **Code Support:** ChatGPT provided guidance on coding structures and library usage (including `libpysal`, `esda`, `spreg`, and `mgwr`), covering spatial weights construction, LISA analysis, GWR bandwidth selection, and spatial regression modeling. It also assisted in troubleshooting and resolving runtime issues encountered during development.
2) **Manuscript Preparation:** ChatGPT was consulted for organizing the report structure and refining section flow. All interpretations, analytical insights, and conclusions were independently developed by the team and validated against the computational results.

All code was reviewed, tested, and validated by the team prior to submission.

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